

Skills Summary

Multiplying & Dividing Fractions

Use this reference while completing your worksheet or when studying.

Skill 1 — Multiply Straight Across

Rule: $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$ — top times top, bottom times bottom.

You may cancel a *top* number with a *bottom* number before multiplying. Never cancel top with top.

Example: Multiply $\frac{3}{5} \times \frac{5}{9}$.

Step 1. This is multiplication, so no flipping — multiply straight across, cancelling a top with a bottom first when you can.

Step 2. Cancel the 5s, and 3 with 9: $\frac{3}{5} \times \frac{5}{9} = \frac{3 \times 5}{5 \times 9} = \frac{15}{45}$

$$\Rightarrow \frac{3}{5} \times \frac{5}{9} = \frac{1}{3}$$

Try it: Multiply $\frac{2}{7} \times \frac{7}{8}$.

check: $\frac{1}{1}$

Skill 2 — Divide: Flip the Second Fraction

Rule (keep-change-flip): $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$

Keep the first fraction, *change* $\div$ to $\times$, *flip* only the *divisor* (the second fraction).

Example: Divide $\frac{4}{5} \div \frac{2}{3}$.

Step 1. Division by a fraction means multiply by its reciprocal — keep the first fraction exactly as it is.

Step 2. Flip only the second: $\frac{4}{5} \times \frac{3}{2} = \frac{12}{10}$, then simplify.

$$\Rightarrow \frac{4}{5} \div \frac{2}{3} = \frac{6}{5}$$

Try it: Divide $\frac{3}{8} \div \frac{3}{4}$.

check: $\frac{2}{1}$

Skill 3 — Spot the Mistake, Then Fix It

Error checklist — when work looks wrong, ask:

- Was the *second* fraction (the divisor) the one that got flipped?
- In a multiply, did the work go *straight across* (not diagonally)?
- Does the answer make sense? Dividing by a fraction smaller than 1 should make the answer *bigger*.

Example: Leo says $\frac{1}{2} \div \frac{1}{6} = \frac{1}{12}$. Find the mistake and the correct answer.

Step 1. Check the answer against sense first: dividing by $\frac{1}{6}$ asks how many sixths fit in one half, so the answer must be bigger than 1 — Leo multiplied instead of flipping the divisor.

Step 2. Fix it: $\frac{1}{2} \times \frac{6}{1} = \frac{6}{2}$
 $\Rightarrow \frac{1}{2} \div \frac{1}{6} = \mathbf{3}$

Try it: Maya says $\frac{3}{4} \div \frac{1}{8} = \frac{3}{32}$. Find her mistake, then compute the correct quotient.

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(!) Watch out: “Flip and multiply” does *not* mean flip whichever fraction you like — only the fraction *after* the $\div$ sign gets flipped, and only in a *division* problem.